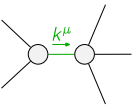
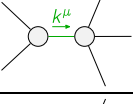
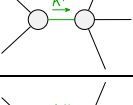
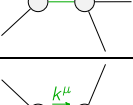
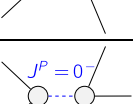
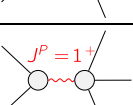
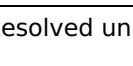


$$\begin{array}{c}
\begin{array}{cc}
\mathcal{K}_{0^+}^{\#1} & \mathcal{K}_{0^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{K}_{0^+}^{\#1}{}^\dagger & \begin{array}{|c|c|} \hline -\frac{\Upsilon_1 k^2}{2} & 0 \\ \hline \end{array} \\
\mathcal{K}_{0^+}^{\#1}{}^{\alpha\beta\chi} & \begin{array}{|c|c|} \hline 0 & \Upsilon_2 k^2 + 3 \frac{(2)}{\kappa_1} \\ \hline \end{array}
\end{array} \\
\begin{array}{cc}
\mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta} & \mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta} & \mathcal{K}_{1^+}^{\#1}{}_{\alpha} & \mathcal{K}_{1^+}^{\#2}{}_{\alpha} \\
\mathcal{K}_{1^+}^{\#1}{}^\dagger{}^{\alpha\beta} & \begin{array}{|c|c|} \hline -\Upsilon_3 k^2 + 2 \frac{(2)}{\kappa_1} & \frac{-\Upsilon_3 k^2 + 2 \frac{(2)}{\kappa_1}}{\sqrt{2}} \\ \hline \end{array} & 0 & 0 \\
\mathcal{K}_{1^+}^{\#2}{}^\dagger{}^{\alpha\beta} & \begin{array}{|c|c|} \hline \frac{-\Upsilon_3 k^2 + 2 \frac{(2)}{\kappa_1}}{\sqrt{2}} & \frac{1}{2} (-\Upsilon_3 k^2 + 2 \frac{(2)}{\kappa_1}) \\ \hline \end{array} & 0 & 0 \\
\mathcal{K}_{1^+}^{\#1}{}^\dagger{}^{\alpha} & 0 & 0 & \frac{\Upsilon_4 k^2}{2} & \frac{\Upsilon_4 k^2}{\sqrt{2}} \\
\mathcal{K}_{1^+}^{\#2}{}^\dagger{}^{\alpha} & 0 & 0 & \frac{\Upsilon_4 k^2}{\sqrt{2}} & \Upsilon_4 k^2
\end{array} \\
\begin{array}{cc}
\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta} & \mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{K}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta} & \begin{array}{|c|c|} \hline \Upsilon_5 k^2 & 0 \\ \hline \end{array} \\
\mathcal{K}_{2^+}^{\#1}{}^{\alpha\beta\chi} & \begin{array}{|c|c|} \hline 0 & \frac{\Upsilon_6 k^2}{2} \\ \hline \end{array}
\end{array}
\end{array}$$

$$\begin{array}{c}
\begin{array}{cc}
\mathcal{J}_{0^+}^{\#1} & \mathcal{J}_{0^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{J}_{0^+}^{\#1}{}^\dagger & \begin{array}{|c|c|} \hline \frac{1}{\text{Det}(0^+)} & 0 \\ \hline \end{array} \\
\mathcal{J}_{0^+}^{\#1}{}^{\alpha\beta\chi} & \begin{array}{|c|c|} \hline 0 & \frac{1}{\text{Det}(0^-)} \\ \hline \end{array}
\end{array} \\
\begin{array}{cc}
\mathcal{J}_{1^+}^{\#1}{}_{\alpha\beta} & \mathcal{J}_{1^+}^{\#2}{}_{\alpha\beta} & \mathcal{J}_{1^+}^{\#1}{}_{\alpha} & \mathcal{J}_{1^+}^{\#2}{}_{\alpha} \\
\mathcal{J}_{1^+}^{\#1}{}^\dagger{}^{\alpha\beta} & \begin{array}{|c|c|} \hline \frac{2}{3 \text{Det}(1^+)} & \frac{\sqrt{2}}{3 \text{Det}(1^+)} \\ \hline \end{array} & 0 & 0 \\
\mathcal{J}_{1^+}^{\#2}{}^\dagger{}^{\alpha\beta} & \begin{array}{|c|c|} \hline \frac{\sqrt{2}}{3 \text{Det}(1^+)} & \frac{1}{3 \text{Det}(1^+)} \\ \hline \end{array} & 0 & 0 \\
\mathcal{J}_{1^+}^{\#1}{}^\dagger{}^{\alpha} & 0 & 0 & \frac{1}{3 \text{Det}(1^-)} & \frac{\sqrt{2}}{3 \text{Det}(1^-)} \\
\mathcal{J}_{1^+}^{\#2}{}^\dagger{}^{\alpha} & 0 & 0 & \frac{\sqrt{2}}{3 \text{Det}(1^-)} & \frac{2}{3 \text{Det}(1^-)}
\end{array} \\
\begin{array}{cc}
\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta} & \mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta\chi} \\
\mathcal{J}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta} & \begin{array}{|c|c|} \hline \frac{1}{\text{Det}(2^+)} & 0 \\ \hline \end{array} \\
\mathcal{J}_{2^+}^{\#1}{}^{\alpha\beta\chi} & \begin{array}{|c|c|} \hline 0 & \frac{1}{\text{Det}(2^-)} \\ \hline \end{array}
\end{array}
\end{array}$$

Abbreviations used in matrices

$$\begin{aligned}
\Upsilon_1 &= 3 \kappa_1^{(4)} - 10 (2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)}) + 21 \kappa_2^{(4)} \&\& \Upsilon_2 = \kappa_{15}^{(4)} - \kappa_{16}^{(4)} \&\& \\
\Upsilon_3 &= 8 \kappa_{15}^{(4)} + 5 \kappa_{16}^{(4)} - 9 \kappa_2^{(4)} + \kappa_3^{(4)} \&\& \Upsilon_4 = 2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)} - 2 \kappa_2^{(4)} \&\& \Upsilon_5 = 10 \kappa_{15}^{(4)} + 5 \kappa_{16}^{(4)} - 9 \kappa_2^{(4)} \&\& \\
\Upsilon_6 &= 2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)} \&\& \text{Det}(0^+) = \frac{1}{2} k^2 (-3 \kappa_1^{(4)} + 20 \kappa_{15}^{(4)} + 10 \kappa_{16}^{(4)} - 21 \kappa_2^{(4)}) \&\& \\
\text{Det}(1^+) &= -\frac{3}{2} k^2 (8 \kappa_{15}^{(4)} + 5 \kappa_{16}^{(4)} - 9 \kappa_2^{(4)} + \kappa_3^{(4)}) + 3 \kappa_1^{(2)} \&\& \text{Det}(2^+) = k^2 (10 \kappa_{15}^{(4)} + 5 \kappa_{16}^{(4)} - 9 \kappa_2^{(4)})
\end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_{1^+}^{\#1\alpha} = \mathcal{J}_{1^+}^{\#2\alpha}$	3	$\partial_\chi \partial^\alpha \mathcal{J}^{\beta\chi}{}_\alpha + \partial_\chi \partial^\chi \mathcal{J}^{\beta\alpha}{}_\beta = 0$	
$\mathcal{J}_{1^+}^{\#1\alpha\beta} = \mathcal{J}_{1^+}^{\#2\alpha\beta}$	3	$3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi} + 2 \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\chi\alpha\beta} = 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi}$	
Total # constraint(s):	6		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$\frac{-22 \kappa_{15}^{(4)} - 11 \kappa_{16}^{(4)} + 18 \kappa_2^{(4)}}{(2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)}) (10 \kappa_{15}^{(4)} + 5 \kappa_{16}^{(4)} - 9 \kappa_2^{(4)})}$
	2	0	$\frac{22 \kappa_{15}^{(4)} + 11 \kappa_{16}^{(4)} - 18 \kappa_2^{(4)}}{(2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)}) (10 \kappa_{15}^{(4)} + 5 \kappa_{16}^{(4)} - 9 \kappa_2^{(4)})}$
	2	0	$\frac{-208 \kappa_{15}^{(4)2} - 208 \kappa_{15}^{(4)} \kappa_{16}^{(4)} - 52 \kappa_{16}^{(4)2} + 372 \kappa_{15}^{(4)} \kappa_2^{(4)} + 186 \kappa_{16}^{(4)} \kappa_2^{(4)} - 171 \kappa_2^{(4)2}}{(2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)}) (20 \kappa_{15}^{(4)2} + 20 \kappa_{15}^{(4)} \kappa_{16}^{(4)} + 5 \kappa_{16}^{(4)2} - 38 \kappa_{15}^{(4)} \kappa_2^{(4)} - 19 \kappa_{16}^{(4)} \kappa_2^{(4)} + 18 \kappa_2^{(4)2})}$
	1	0	$-\frac{16 \kappa_{15}^{(4)2} + 4 \kappa_{16}^{(4)2} + 16 \kappa_{15}^{(4)} (\kappa_{16}^{(4)} - \kappa_2^{(4)}) - 8 \kappa_{16}^{(4)} \kappa_2^{(4)} + \sqrt{(2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)})^2 (452 \kappa_{15}^{(4)2} + 452 \kappa_{15}^{(4)} \kappa_{16}^{(4)} + 113 \kappa_{16}^{(4)2} - 832 \kappa_{15}^{(4)} \kappa_2^{(4)} - 416 \kappa_{16}^{(4)} \kappa_2^{(4)} + 384 \kappa_2^{(4)2})}}{(2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)}) (20 \kappa_{15}^{(4)2} + 20 \kappa_{15}^{(4)} \kappa_{16}^{(4)} + 5 \kappa_{16}^{(4)2} - 38 \kappa_{15}^{(4)} \kappa_2^{(4)} - 19 \kappa_{16}^{(4)} \kappa_2^{(4)} + 18 \kappa_2^{(4)2})}$
	1	0	$-\frac{16 \kappa_{15}^{(4)2} - 4 \kappa_{16}^{(4)2} - 16 \kappa_{15}^{(4)} (\kappa_{16}^{(4)} - \kappa_2^{(4)}) + 8 \kappa_{16}^{(4)} \kappa_2^{(4)} + \sqrt{(2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)})^2 (452 \kappa_{15}^{(4)2} + 452 \kappa_{15}^{(4)} \kappa_{16}^{(4)} + 113 \kappa_{16}^{(4)2} - 832 \kappa_{15}^{(4)} \kappa_2^{(4)} - 416 \kappa_{16}^{(4)} \kappa_2^{(4)} + 384 \kappa_2^{(4)2})}}{(2 \kappa_{15}^{(4)} + \kappa_{16}^{(4)}) (20 \kappa_{15}^{(4)2} + 20 \kappa_{15}^{(4)} \kappa_{16}^{(4)} + 5 \kappa_{16}^{(4)2} - 38 \kappa_{15}^{(4)} \kappa_2^{(4)} - 19 \kappa_{16}^{(4)} \kappa_2^{(4)} + 18 \kappa_2^{(4)2})}$
	1	$\frac{3 \kappa_1^{(2)}}{-\kappa_{15}^{(4)} + \kappa_{16}^{(4)}}$	$\frac{1}{-\kappa_{15}^{(4)} + \kappa_{16}^{(4)}}$
	3	$\frac{2 \kappa_1^{(2)}}{8 \kappa_{15}^{(4)} + 5 \kappa_{16}^{(4)} - 9 \kappa_2^{(4)} + \kappa_3^{(4)}}$	$-\frac{2}{3 (8 \kappa_{15}^{(4)} + 5 \kappa_{16}^{(4)} - 9 \kappa_2^{(4)} + \kappa_3^{(4)})}$
Resolved unitarity condition(s):		(Demonstrably impossible)	